

知识追踪与多模态认知让渡研究文献综述

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摘要

本文档由本地文献精读 workflow 自动生成，汇总 2026 年度知识追踪（Knowledge Tracing）与多模态认知让渡检测两个方向共 46 篇代表性文献。全文按四个主题组织为独立综述章节，每篇文献附原文摘录、中文翻译与引用条目；另设专章总结 10 篇未能获取全文的重要文献。

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1 知识追踪综述

本章汇总知识追踪（Knowledge Tracing）领域的综述性研究，涵盖深度学习、图神经网络等模型谱系、主流变体与典型应用场景，为后续研究提供全景式参照。

1.1 A Survey of Knowledge Tracing: Models, Variants, and Applications

引用: [31]

Knowledge Tracing (KT) is one of the fundamental tasks for student behavioral data analysis, aiming to monitor students' evolving knowledge state during their problem-solving process.

译文: 知识追踪（KT）是学生行为数据分析的基本任务之一，旨在监测学生在问题解决过程中不断演化的知识状态。

In recent years, a substantial number of studies have concentrated on this rapidly growing field, significantly contributing to its advancements.

译文: 近年来，大量研究聚焦于这一快速增长的领域，有力地推动了它的发展。

In this survey, we will conduct a thorough investigation of these progressions.

译文: 本综述将对这些进展进行全面而深入的考察。

1.2 Knowledge Tracing: A Survey

引用: [1]

A human teacher can track the knowledge of students to customize the teaching on students' needs.

译文: 人类教师能够掌握学生的知识状况，从而根据学生的需求定制教学。

With the rise of online education platforms, there is a similar need for machines to track the knowledge of students and tailor their learning experience.

译文: 随着在线教育平台的兴起，机器也同样需要追踪学生的知识水平并为其量身定制学习体验。

This is known as the Knowledge Tracing (KT) problem in the literature.

译文: 这一问题在文献中被称为知识追踪（KT）问题。

1.3 A Survey of Graph-Based Knowledge Tracing from Structural and Propagation Perspectives: Models, Applications, and Future Directions

引用: [37]

Early deep learning-based KT mainly relies on sequential models, which often find it more challenging to explicitly model the complex many-to-many relationships between students and knowledge entities, such as exercises and knowledge components.

译文: 早期的深度知识追踪主要依赖序列模型, 这类模型往往难以显式地建模学生与习题、知识组件等知识实体之间复杂的多对多关系。

Graph-based Knowledge Tracing (GbKT) further leverages graph neural networks to explicitly represent these structural relationships, creating new opportunities for learning-state monitoring and personalized intervention.

译文: 基于图的知识追踪 (GbKT) 进一步利用图神经网络显式表示这些结构关系, 为学习状态监测和个性化干预创造了新的机会。

In this survey, we conduct an in-depth exploration of recent progress in GbKT research.

译文: 本综述对 GbKT 研究的最新进展进行了深入探讨。

1.4 Survey on Deep Knowledge Tracing Methods Based on Graph Neural Networks

引用: [42]

基于图神经网络的深度知识追踪方法通过图结构有效建模学习者与知识点的复杂关系, 克服了传统模型的局限性, 展现出更高的预测精度和更强的可解释性。

总结基于图神经网络、图卷积网络、图注意力网络和图记忆网络的深度知识追踪方法, 并从采用不同图结构、增加额外特征、融合教育心理学因素三个角度对相关研究工作进行分类。

1.5 Knowledge Tracing Models in Educational Data Mining: Historical Evolution, Categorization, and Empirical Evaluation

引用: [6]

This article analyses computational models of Knowledge Tracing (KT), which address the complex sequence-modelling task of predicting dynamic, unobservable latent user states from historical interaction logs.

译文: 本文分析知识追踪 (KT) 的计算模型, 这些模型处理的是从历史交互日志中预测动态的、不可观测的潜在用户状态这一复杂的序列建模任务。

First, we propose a comprehensive taxonomy identifying nine distinct and interconnected KT model families: psychometric; Bayesian; machine learning; deep learning; graph-based; temporal/sequential; multi-task; contrastive/self-supervised; and domain-adaptive.

译文: 首先, 我们提出了一个全面的分类体系, 识别出九个相互关联的 KT 模型家族: 心理测量、贝叶斯、机器学习、深度学习、基于图、时序/序列、多任务、对比/自监督以及领域自适应。

Secondly, we trace the historical evolution of KT architectures, from the foundational psychometric methods of the 1950s to the modern integration of attention mechanisms and graph neural networks.

译文: 其次, 我们追溯了 KT 架构的历史演进, 从 20 世纪 50 年代的基础心理测量方法, 一直到注意力机制与图神经网络的现代融合。

2 知识追踪教育实践

本章聚焦知识追踪在真实教育场景中的落地工作，包括可解释建模、隐私遗忘学习、RNN 轻量化建模与领域专属智能教学系统等方向。

2.1 UnlearnKT: Machine Unlearning for Privacy-Preserving Knowledge Tracing in E-Learning IoT Systems

引用: [12]

However, KT models may retain information about specific learners even after their data has been removed, creating privacy and compliance risks.

译文: 然而, 即使学习者的数据已被删除, KT 模型仍可能保留关于特定学习者的信息, 从而带来隐私与合规风险。

To address this problem, we first present UnlearnKT, a KT-specific machine unlearning framework that organizes learner-level deletion requests into standard training, targeted unlearning, and post-unlearning evaluation.

译文: 为解决这一问题, 我们首先提出 UnlearnKT——一个专门面向 KT 的机器遗忘框架, 将学习者级别的删除请求组织为标准训练、定向遗忘和遗忘后评估三个阶段。

These results show that QEFU-KT offers the strongest overall quality-efficiency trade-off for privacy-preserving KT in E-IoT systems.

译文: 结果表明, QEFU-KT 在 E-IoT 系统的隐私保护知识追踪中提供了整体质量-效率权衡上的最优表现。

2.2 Towards Student Assessment: Last-Query Transformer with RNN for Efficient Long-Term Knowledge Tracing

引用: [19]

However, a fundamental challenge remains: effectively modeling students' long-term learning trajectories to support comprehensive assessment and targeted instruction.

译文: 然而, 一个根本性的挑战依然存在: 如何有效建模学生的长期学习轨迹, 以支持全面评估和针对性教学。

Existing approaches face significant limitations in processing extended learning sequences due to computational constraints and insufficient modeling of temporal dependencies.

译文：受计算资源约束以及对时间依赖关系建模不足的影响，现有方法在处理长学习序列时存在明显局限。

This paper presents an efficient deep learning framework for long-term student modeling that combines the global relationship learning of Transformers with the temporal dynamics of Recurrent Neural Networks.

译文：本文提出了一种高效的长期学生建模深度学习框架，将 Transformer 的全局关系学习能力与循环神经网络的时间动态建模能力相结合。

2.3 Dynamic Heterogeneous Graph-Structured Knowledge Tracing with Multi-Behavior Integration and Temporal Forgetting

引用：[29]

Most existing KT methods focus on exercise-response sequences and therefore cannot explicitly distinguish knowledge exposure through learning resources from knowledge verification.

译文：现有大多数 KT 方法只关注「练习—作答」序列，因而无法明确区分通过学习资源获得的知识暴露与通过作答完成的知识检验。

We propose Dynamic Heterogeneous Graph-Structured Knowledge Tracing (DHGKT), a framework that organizes learners, exercises, videos, concepts, and optional comments within a unified dynamic heterogeneous graph.

译文：我们提出动态异构图结构知识追踪（DHGKT）框架，将学习者、习题、视频、知识点以及可选的评论关系统一组织在一张动态异构图中。

These results indicate that the principal value of DHGKT lies in transparent event-driven knowledge-state evolution rather than generic graph convolution.

译文：这些结果表明，DHGKT 的主要价值在于透明的事件驱动式知识状态演化，而非通用的图卷积。

2.4 Exploring Interpretable Deep Knowledge Tracing Through Integrating SHAP with LLMs

引用：[28]

Deep Knowledge Tracing (DKT) advances KT through deep learning techniques applied to large-scale educational datasets, enabling the discovery of latent patterns from the education data.

译文：深度知识追踪（DKT）将深度学习技术应用于大规模教育数据集，能够从教育数据中发现潜在模式，从而推动 KT 的发展。

However, the black-box nature of DKT models limits their adoption in real-world educational settings, where interpretability and trust are critical.

译文：然而，DKT 模型的黑箱特性限制了它在真实教育场景中的落地，因为这些场景对可解释性和可信度至关重要。

To address this challenge, we propose a novel framework that enhances the interpretability of DKT by integrating SHapley Additive exPlanations (SHAP) to quantify the contribution of individual features.

译文：为应对这一挑战，我们提出了一个新框架，通过整合 SHAP（沙普利加性解释）量化各个特征的贡献，从而增强 DKT 的可解释性。

2.5 Application and Effectiveness Analysis of AI Intelligent Tutoring System Combined with Knowledge Graph in Basic Medical Pathology Teaching

引用：[39]

Background: Basic medical pathology teaching suffers from fragmented knowledge and insufficient personalized learning, which impairs students' systematic knowledge construction and learning efficiency.

译文：背景：基础医学病理学教学存在知识碎片化、个性化学习不足的问题，影响了学生系统性的知识建构和学习效果。

This study aimed to apply an AI and knowledge graph integrated teaching model to address these challenges and provide references for basic medical education reform.

译文：本研究旨在应用「AI + 知识图谱」融合教学模式来应对这些挑战，为基础医学教育改革提供参考。

Methods: An 18-week randomized controlled trial was conducted on 360 nursing students (180 each in experimental/control groups).

译文：方法：研究对 360 名护理专业学生开展了为期 18 周的随机对照试验（试验组与对照组各 180 人）。

2.6 Convergence Patterns between Knowledge Tracing and Knowledge Graphs in Technology-Enhanced Learning

引用: [27]

This study examines the relationship and emerging convergence patterns between Knowledge Tracing (KT) and Knowledge Graphs (KG) within Technology-Enhanced Learning (TEL) through a bibliometric and thematic analysis of their evolution.

译文: 本研究通过文献计量与主题分析, 考察技术增强学习 (TEL) 领域中知识追踪 (KT) 与知识图谱 (KG) 之间的关联及其新兴的融合趋势。

The analysis was based on 321 Scopus-indexed documents published between 2015 and 2025 and was conducted using VOSviewer and Biblioshiny (R-Bibliometrix) to identify publication trends, conceptual structures, and thematic evolution.

译文: 分析基于 2015 - 2025 年间发表的 321 篇 Scopus 收录文献, 使用 VOSviewer 和 Biblioshiny (R-Bibliometrix) 识别发表趋势、概念结构与主题演化。

Although recent literature increasingly references connections between KT and KG, the corpus reveals that publications predominantly fall into KT-only and KG-only domains, with a subset exhibiting conceptual intersections rather than explicit methodological integration.

译文: 尽管近期文献越来越多地提及 KT 与 KG 的关联, 但语料显示出版物主要仍分属纯 KT 或纯 KG 领域, 仅有一小部分呈现概念上的交叉, 而并非明确的方法层面融合。

3 多模态认知让渡检测

本章围绕生成式人工智能时代学习者的认知卸载/让渡现象，汇总多模态数据驱动的检测、量表开发与影响因素研究。

3.1 AI tools in society: Impacts on cognitive offloading and the future of critical thinking

引用: [10]

This study investigates the relationship between AI tool usage and critical thinking skills, focusing on cognitive offloading as a mediating factor.

译文: 本研究考察 AI 工具使用与批判性思维能力之间的关系，重点将认知让渡作为中介因素。

Utilising a mixed-method approach, we conducted surveys and in-depth interviews with 666 participants across diverse age groups and educational backgrounds.

译文: 研究采用混合方法，对来自不同年龄组和教育背景的 666 名参与者进行了问卷调查与深度访谈。

3.2 Generative AI tool use enhances academic achievement in sustainable education through shared metacognition and cognitive offloading among preservice teachers

引用: [15]

Therefore, the present study investigates the connection between generative artificial intelligence tool usage (GenAITU) and academic achievement (AA) in the context of SDG4.

译文: 因此，本研究在可持续发展目标 4 (SDG4) 的背景下考察生成式人工智能工具使用 (GenAITU) 与学业成绩 (AA) 之间的关联。

We assessed the mediating role of shared metacognition (SMC) and cognitive offloading (COL) in this relationship among preservice teachers (PSTs).

译文: 我们评估了在职前教师 (PST) 群体中共享元认知 (SMC) 与认知让渡 (COL) 在该关系中所起的中介作用。

The authors surveyed 465 students from five universities in Wuhan, China, using a 7-point Likert scale through a time-lag design.

译文: 作者采用时间滞后设计，使用 7 点李克特量表对中国武汉五所高校的

465 名学生进行了调查。

3.3 Not all cognitive offloading is equal: distinguishing dependent and autonomous offloading to generative AI

引用: [44]

Yet research on cognitive offloading has largely treated AI use as a unidimensional phenomenon, obscuring a theoretically consequential distinction: whether AI substitutes for the user's own thinking or merely assists it.

译文: 然而, 认知让渡研究大多将 AI 使用视为单维现象, 掩盖了一个在理论上具有重要意义的区分: AI 究竟是替代了使用者自身的思考, 还是仅仅辅助思考。

Drawing on the autonomous-dependent help typology and self-determination theory, the present study introduces and examines a distinction between dependent cognitive offloading (delegating core thinking) and autonomous cognitive offloading (using AI to support one's own thinking).

译文: 本研究借鉴自主—依赖型帮助类型学与自我决定理论, 引入并考察了依赖型认知让渡(将核心思考委托给 AI)与自主型认知让渡(利用 AI 支撑自身思考)之间的区分。

Dependent offloading was positively associated with cognitive agency transfer and negatively associated with intrinsic motivation, which in turn were linked to poorer perceived outcomes.

译文: 依赖型让渡与认知主体性转移呈正相关、与内在动机呈负相关, 而这两者又与更差的主观结果相关联。

3.4 CLARE: Cognitive load assessment in real-time with multimodal data

引用: [4]

We present a novel multimodal dataset for Cognitive Load Assessment in REal-time (CLARE).

译文: 我们提出了一个用于实时认知负荷评估的新型多模态数据集 CLARE。The dataset contains physiological and gaze data from 24 participants with self-reported cognitive load scores as ground-truth labels.

译文: 该数据集包含 24 名参与者的生理信号与眼动数据, 并以自评认知负

荷分数作为真值标签。

3.5 Artificial intelligence, cognitive offloading and implications for education

引用: [20]

This report forms part of the work program supporting the Australian Network for Quality Digital Education.

译文: 本报告是澳大利亚优质数字教育网络工作计划的一部分。

Members of the Network have provided valuable engagement, input and feedback as part of the report's development, though the report does not represent a consensus or endorsed Network view.

译文: 网络成员在报告撰写过程中提供了宝贵的参与、意见与反馈, 但本报告并不代表该网络的共识或认可观点。

3.6 Pedagogical partnerships with generative AI in higher education: how dual cognitive pathways paradoxically enable transformative learning

引用: [36]

While GenAI offers unprecedented speed in knowledge processing, it raises concerns that efficiency-driven motivations may erode the cognitive struggle essential for intellectual growth.

译文: 尽管生成式 AI 在知识处理上提供了前所未有的速度, 但也引发了担忧: 效率驱动的动机可能会侵蚀智力成长所必需的认知挣扎。

This study investigates how Human-GenAI pedagogical partnerships influence transformative learning through the dual cognitive pathways of vigilance and offloading, moderated by efficiency orientation.

译文: 本研究考察人机—生成式 AI 教学伙伴关系如何通过警觉与让渡双重认知路径影响变革性学习, 并探讨效率导向的调节作用。

Contrary to traditional assumptions that delegation diminishes learning depth, our results indicate that strategic cognitive offloading—when exceeding specific thresholds—liberates mental resources for transformative learning.

译文: 与「委托会削弱学习深度」的传统假设相反, 我们的结果表明, 策略

性认知让渡在超过特定阈值时能够释放心智资源，服务于变革性学习。

3.7 ...thinking is outsourced: Cognitive offloading and the heterogeneity of critical thinking among Chinese university students using generative artificial intelligence

引用: [32]

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...thinking is outsourced: Cognitive offloading and the heterogeneity of critical thinking among Chinese university students using generative artificial intelligence

3.8 Academic Dependency, AI Literacy, and Cognitive Offloading Predict Students' Cognitive Ability in Generative AI Learning

引用: [9]

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Academic Dependency, AI Literacy, and Cognitive Offloading Predict Students' Cognitive Ability in Generative AI Learning

3.9 Mediating role of cognitive offloading: linkage between generative AI usage and critical thinking

引用: [3]

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Mediating role of cognitive offloading: linkage between generative AI usage and critical thinking

3.10 Teachers as reflective regulators of cognition: Understanding cognitive offloading in AI-augmented practice

引用: [13]

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Teachers as reflective regulators of cognition: Understanding cognitive offloading in AI-augmented practice

3.11 The silent skill erosion: Cognitive offloading in the age of educational AI

引用: [40]

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The silent skill erosion: Cognitive offloading in the age of educational AI

3.12 Development and psychometric validation of the brain rot scale: measuring digital content overconsumption among generation alpha and generation Z

引用: [26]

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Development and psychometric validation of the brain rot scale: measuring digital content overconsumption among generation alpha and generation Z

3.13 Artificial intelligence threatens critical thinking in education systems

引用: [46]

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Artificial intelligence threatens critical thinking in education systems

3.14 Cognitive Offloading and Autonomous Agency: A Comprehensive Review of Self-Regulating Smartphone Architectures

引用: [34]

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Cognitive Offloading and Autonomous Agency: A Comprehensive Review of Self-Regulating Smartphone Architectures

3.15 Cognitive offloading and the reshaping of human thought: The subtle influence of Artificial Intelligence

引用: [14]

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Cognitive offloading and the reshaping of human thought: The subtle influence of Artificial Intelligence

3.16 In-Class AI Use and Attitudes Among University Students: The Different Mediating Roles of Cognitive Relief and Cognitive Offloading

引用: [8]

(注: 该文 PDF 原文未能成功下载, 此处以题录信息代替原文摘录; 相关工作总结见第五章。)

In-Class AI Use and Attitudes Among University Students: The Different Mediating Roles of Cognitive Relief and Cognitive Offloading

3.17 Behavioral Indicators of Overreliance During Interaction with Conversational Language Models

引用: [17]

However, their inherent hallucinations risk hiding misinformation in fluent responses, raising concerns about overreliance on AI.

译文: 然而, LLM 固有的幻觉问题可能把错误信息隐藏在流畅的回复中, 引发了对过度依赖 AI 的担忧。

Detecting overreliance is challenging, as it often arises in complex, dynamic contexts and cannot be easily captured by post-hoc task outcomes.

译文: 检测过度依赖十分困难, 因为它往往出现在复杂、动态的情境中, 无法通过事后任务结果轻易捕捉。

We collected interaction logs from 77 participants working with an LLM injected plausible misinformation across three real-world tasks and we assessed overreliance by whether participants detected and corrected these errors.

译文: 我们采集了 77 名参与者在三个真实任务中与注入了貌似合理错误信息的 LLM 交互的日志, 并通过参与者是否发现并纠正这些错误来评估过度依赖。

By semantically encoding and clustering segments of user interactions, we identified five behavioral patterns linked to overreliance.

译文: 通过对用户交互片段进行语义编码与聚类, 我们识别出五种与过度依赖相关的行为模式。

4 AI 教育应用及其他

本章收录与知识追踪相邻的 AI 教育应用研究, 包括自动作文评分、教育数据挖掘、数学推理评测与 K12 AI 教育综述等方向。

4.1 Systematic review of research on artificial intelligence in K-12 education (2017–2022)

引用: [24]

Purpose: The purpose of this systematic review is to provide a more complete and nuanced understanding of the role and impact of AI in K-12 education by synthesizing publication trends, AI research themes, AI methods and technology applications, and AI use by students and teachers in K-12 educational settings.

译文: 目的: 本系统综述通过综合 K-12 教育场景中的发表趋势、AI 研究主题、AI 方法与技术应用以及师生对 AI 的使用情况, 更全面、细致地理解 AI 在 K-12 教育中的作用与影响。

Methods: The systematic review searched Web of Science and six databases indexed in EBSCO host.

译文: 方法: 本系统综述检索了 Web of Science 以及 EBSCO host 收录的六个数据库。

4.2 Educational data mining and learning analytics: An updated survey

引用: [30]

This survey is an updated and improved version of the previous one published in 2013 in this journal with the title Data Mining in Education.

译文: 本综述是 2013 年发表于本刊的《教育中的数据挖掘》一文的更新与改进版。

This paper provides the current state of the art by reviewing the main publications, the key milestones, the knowledge discovery cycle, the main educational environments, the specific tools, the free available datasets, the most used methods, the main objectives and the future trends in this research area.

译文: 本文通过梳理主要出版物、关键里程碑、知识发现循环、主要教育环境、专用工具、公开可用的数据集、最常用的方法、主要目标以及未来趋势,

给出该研究领域的最新研究现状。

4.3 Automated Essay Scoring: A Survey of the State of the Art

引用: [16]

Automated essay scoring (AES), the task of employing computer technology to score written text, is one of the most important educational applications of natural language processing (NLP).

译文: 自动作文评分 (AES) 即利用计算机技术为书面文本打分, 是自然语言处理 (NLP) 最重要的教育应用之一。

This area of research began with Page's [1966] pioneering work on the Project Essay Grader system and has remained active since then.

译文: 该研究领域始于 Page (1966) 关于 Project Essay Grader 系统的开创性工作, 此后一直持续发展。

4.4 A survey on deep learning-based automated essay scoring and feedback generation

引用: [25]

However, these models face a critical limitation in explaining the specific patterns and features utilized for scoring, which are essential for interpreting the scores and offering constructive feedback to essay authors.

译文: 然而, 这些模型在解释评分所依据的具体模式与特征方面存在关键局限, 而这些解释对于理解分数并向作者提供建设性反馈至关重要。

While existing surveys on AES cover topics ranging from representation to scoring models, they primarily emphasize scoring models.

译文: 现有的 AES 综述虽然涵盖从表示方法到评分模型等多个主题, 但主要侧重于评分模型。

4.5 Survey on Automated Short Answer Grading with Deep Learning: from Word Embeddings to Transformers

引用: [11]

Recent progress in Natural Language Processing and Machine Learning has largely influenced the field of ASAG, of which we survey the recent research advancements.

译文:自然语言处理与机器学习的最新进展深刻影响了自动短文评分(ASAG)领域, 本文对其最新研究进展进行综述。

In particular, we focus our analysis on the transition from hand-engineered features to representation learning approaches, which learn representative features for the task at hand automatically from large corpora of data.

译文:特别地, 我们重点分析从手工设计特征向表示学习方法的转变——后者能够从大规模语料中自动学习当前任务的有效特征。

4.6 The Gap of Semantic Parsing: A Survey on Automatic Math Word Problem Solvers

引用: [41]

Despite the long history dated back to the 1960s, MWPs have regained intensive attention in the past few years with the advancement of Artificial Intelligence (AI).

译文: 尽管数学应用题(MWP)的研究历史可以追溯到20世纪60年代, 但随着人工智能(AI)的发展, 该方向在过去几年重新受到密集关注。

Many systems have claimed promising results in self-crafted and small-scale data sets.

译文: 许多系统宣称在自编数据集和小规模数据集上取得了有希望的结果。

4.7 A Survey of Deep Learning for Mathematical Reasoning

引用: [22]

The development of artificial intelligence (AI) systems capable of solving math problems and proving theorems in language has garnered significant interest in the fields of machine learning and natural language processing.

译文: 能够用语言解决数学问题和证明定理的人工智能(AI)系统的开发, 在机器学习与自然语言处理领域引起了广泛关注。

On the other hand, recent advances in large-scale neural language models have opened up new benchmarks and opportunities to use deep learning for mathematical reasoning.

译文：另一方面，大规模神经语言模型的最新进展，为利用深度学习进行数学推理开辟了新的基准与机会。

4.8 Review on Neural Question Generation for Education Purposes

引用：[2]

However, creating high-level questions requires significant creativity and effort.

译文：然而，创建高水平的问题需要相当的创造力与精力投入。

Automatic question generation is expected to facilitate the generation of not only fluent and relevant but also educationally valuable questions.

译文：自动问题生成有望促进生成不仅流畅、相关，而且具有教育价值的问题。

While rule-based methods are intuitive for short inputs, they struggle with longer and more complex inputs.

译文：基于规则的方法对短输入来说较为直观，但难以应对更长、更复杂的输入。

5 未能获取全文的文献工作综述

以下 10 篇文献因出版方平台限制（SAGE、ScienceDirect、IEEE、ACM、Taylor & Francis 等反爬或付费墙）未能下载 PDF/TeX 原文，但其主题与本综述高度相关，故依据检索题录与本地学习笔记对其工作做摘要性总结；详细题录见参考文献中相应条目（备注字段已注明）。

5.1 song2022survey

引用：[33] 深度学习知识追踪综述。围绕序列建模、记忆增强、注意力机制、知识结构与多模态/辅助信息梳理 DKT 之后的模型演进路线，归纳出可解释性、长程依赖和特征不足三类开放挑战，将相关研究组织为一张可检索、可比较的领域地图，是进入深度 KT 研究的基础读物。

5.2 zhu2026dkpkt

引用：[45] 时序知识追踪方法论文。提出 DKPKT 框架，通过交互感知模块（动态注意力）与知识感知模块（遗忘/获得因子）联合建模学生知识状态的动态演化，显式区分「遗忘」与「获得」两种相反机制，在三个真实教育数据集上超过 11 个主流基线，对时序 KT 建模具有参考价值。

5.3 dizon2026assessing

引用：[7] 量表开发研究。针对生成式 AI 时代学生将目标设定、错误监控、策略反思等元认知活动也外包给 AI 的新现象，开发并初步验证了 6 条目的「元认知懒惰量表」(MLS)：内部一致性优秀 ($\alpha = .95$)，元认知懒惰能显著预测学习疏离（行为疏离 $\beta = .49$ 、情感疏离 $\beta = .50$ ）而与学习投入无显著关系，将认知让渡研究从「任务外包」推进到「元认知外包」层面。

5.4 j2018multimodal

引用：[43] 手册综述章节（ACM 多模态交互手册）。系统梳理行为信号（眼动、面部表情、头部姿态、交互行为）与生理信号（EEG、ECG、皮电、呼吸等）作为认知负荷指标的证据、特征与融合方法，是该领域早期权威综述之一；其出发点是传统主观量表（NASA-TLX）与绩效指标无法实时、客观地测量认知负荷，多模态传感为无扰式连续测量提供可能。

5.5 j2025learners

引用: [35] 实证研究 (被引 209 次)。对 580 名中国大学生的调节中介模型 (PROCESS Model 8) 分析显示: AI 依赖越深, 批判性思维越低, 认知疲劳是部分中介机制; AI 素养缓冲 AI 依赖对批判性思维的直接损害, 但在高依赖时会放大认知疲劳, 呈现「双刃剑」效应。该文章将「AI 依赖」确立为与认知让渡并行的新构念。

5.6 jiang2026ai-based

引用:[38] 量表开发研究。针对中国高校学生开发并初步验证了「AI 认知让渡量表」(AIBCO): 5 点 Likert 频率量表, 捕捉学习者把低阶与高阶认知任务都外包给 AI 的倾向, 填补了中国高等教育语境下 AI 让渡测量的工具空白。

5.7 liu2023cognitive

引用: [18] 多模态负荷估计的代表性深度学习工作 (IEEE JBHI, 被引 58 次)。提出多视图学习框架, 融合 EEG、EDA、ECG、EOG、眼动五类生理模态预测认知负荷, 核心结论「融合的价值来自互补性而非传感器数量」被 2026 年多篇综述引用, 对本课题的多模态融合策略有直接借鉴意义。

5.8 lodge2023calculator

引用: [21] 概念性评论 (被引 170+, 认知让渡教育讨论的基础文献)。指出计算器只接管计算、把「判断算什么」留给用户, 而生成式 AI 能接管总结、分析、创作、评价——即学习过程本身的认知操作, 因此「AI 只是工具」的隐喻失效, 学习者与 AI 的关系需要重新概念化。

5.9 m2019labelless

引用: [5] 方法论文 (ESI 高被引 125 次)。提出「无标签学习」框架: 利用源域有标签数据与目标域无标签数据之间的领域适配/对抗学习, 使基于 EEG 等生理信号的情绪认知摆脱对目标场景大量人工标注的依赖; 其跨被试/跨设备迁移思路对多模态认知让渡检测的标注瓶颈有直接借鉴意义。

5.10 mohamed2026hybrid

引用: [23] 框架设计研究。提出人机混合 (Hybrid Human-AI) 的调控学习框架, 用因果推断识别「何时发生认知让渡」, 从而在生成式 AI 教育环境中主动缓解过度让渡; 其未来扩展方向明确包括基于生理信号的多模态让渡检测, 与本课题方向高度相关。

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